Exercise 1.

(a) $E(X) = \sum_{i=1}^{n} P(X = x_i)x_i$

$E(X) = \sum_{i=1}^{n} \frac{1}{n}x_i$

$E(X) = \mu = \frac{1}{n}\sum_{i=1}^{n} x_i$

$Var(X) = E(X^2) - E(X)^2$

$E(X^2) = \sum_{i=1}^{n} P(X = x_i){x_i}^2 = \frac{1}{n}\sum_{i=1}^{n} {x_i}^2$

$Var(X) = \frac{1}{n}\sum_{i=1}^{n} {x_i}^2 - (\frac{1}{n}\sum_{i=1}^{n} x_i)^2$

(b) $E(X) = \frac{1}{n}\sum_{k=1}^{n} k$

$E(X) = \frac{1}{n}\frac{n(n+1)}{2}$

$E(X) = \frac{(n+1)}{2}$

$Var(X) = E(X^2) - E(X)^2$

$E(X^2) = \frac{1}{n}\sum_{k=1}^{n} k^2$

$E(X^2) = \frac{1}{n}\frac{n(n+1)(2n+1)}{6}$

$E(X^2) = \frac{(n+1)(2n+1)}{6}$

$Var(X) = \frac{(n+1)(2n+1)}{6} - (\frac{(n+1)}{2})^2$

$Var(X) = \frac{2n^2+n+2n+1}{6} - \frac{(n+1)^2}{4}$

$Var(X) = \frac{2n^2+3n+1}{6} - \frac{n^2+2n+1}{4}$

$Var(X) = \frac{2(2n^2+3n+1)-3(n^2+2n+1)}{12}$

$Var(X) = \frac{4n^2+6n+2-3n^2-6n-3}{12}$

$Var(X) = \frac{(n^2-1)}{12}$

Exercise 2.

show: $Var(aX + b) = a^2Var(X)$

$Var(aX + b) = E[((aX + b) - E(aX + b))^2]$

$= E[((aX + b) - (a\mu + b)^2]$

$= E[(aX - a\mu + b - b)^2]$

$= E[(a(x - \mu))^2]$

$= E[a^2(x - \mu)^2]$

$= a^2Var(X)$

Exercise 3.

(a) mean of X - Poisson(λ)

X - Poisson(λ) $= e^{-\lambda}\frac{\lambda^x}{x!}$ x = 0, 1, 2, 3, ...

$E(X) = \sum^X xe^{-\lambda}\frac{\lambda^x}{x!}$

$E(X) = \sum_0^\infty xe^{-\lambda}\frac{\lambda^x}{x!}$

$E(X) = \sum_1^\infty xe^{-\lambda}\frac{\lambda^x}{x!}$

$E(X) = e^{-\lambda}\sum_1^\infty \lambda^x\frac{x}{x!}$

$E(X) = e^{-\lambda}\sum_1^\infty \frac{\lambda^x}{(x-1)!}$

$E(X) = e^{-\lambda}\lambda\sum_1^\infty \frac{\lambda^{x-1}}{(x-1)!}$

$u = x - 1, u + 1 = x$

$E(X) = e^{-\lambda}\lambda\sum_0^\infty \frac{\lambda^u}{(u)!}$

$E(X) = \lambda e^{-\lambda}e^u$

$E(X) = \lambda e^{-\lambda}e^\lambda$

$E(X) = \lambda$

(b) $E(X^2) = \sum_X x^2e^{-\lambda}\frac{\lambda^x}{x!}$

$= \sum_{x=0}^\infty x^2e^{-\lambda}\frac{\lambda^x}{x!}$

$= \sum_{x=1}^\infty x^2e^{-\lambda}\frac{\lambda^x}{x!}$

$= \sum_{x=1}^\infty xe^{-\lambda}\frac{\lambda^x}{(x-1)!}$

$= \sum_{x=1}^\infty((x-1)+1)e^{-\lambda}\frac{\lambda^x}{(x-1)!}$

$= \sum_{x=1}^{\infty}(x-1)e^{-\lambda}\frac{\lambda^x}{(x-1)!} + \sum_{x=1}^{\infty} e^{-\lambda}\frac{\lambda^x}{(x-1)!}$

$= (1-1)e^{-1}\frac{\lambda}{(1-1)!} + \sum_{x=2}^{\infty} e^{-\lambda}\frac{\lambda^x}{(x-2)!} + \sum_{x=1}^{\infty} e^{-\lambda}\frac{\lambda^x}{(x-1)!}$

$= 0 + e^{-\lambda}\sum_{x=2}^{\infty}\frac{\lambda^x}{(x-2)!} + e^{-\lambda}\sum_{x=1}^{\infty}\frac{\lambda^x}{(x-1)!}$

$= e^{-\lambda}\lambda^2\sum_{x=2}^{\infty}\frac{\lambda^{x-2}}{(x-2)!} + e^{-\lambda}\lambda\sum_{x=1}^{\infty}\frac{\lambda^{x-1}}{(x-1)!}$

$u = x-2, a = x-1$

$u = u+2, x = a+1$

$= e^{-\lambda}\lambda^2\sum_{x=2}^{\infty}\frac{\lambda^u}{u!} + e^{-\lambda}\lambda\sum_{x=1}^{\infty}\frac{\lambda^a}{a!}$

$= e^{-\lambda}\lambda^2 e^{\lambda} + e^{-\lambda}\lambda e^{-\lambda}$

$= \lambda^2 + \lambda$

(c) $M(\theta) = E(e^{\theta x})$

$M(\theta) = E(e^{\theta x}) = \sum_{x=0}^{\infty} e^{\theta x}e^{-\lambda}\frac{\lambda^x}{x!}$

$= e^{-\lambda}\sum_{x=0}^{\infty}\frac{(\lambda e^{\theta})^x}{x!}$

$= e^{-\lambda}e^{\lambda e^{\theta}}$

$= e^{\lambda e^{\theta} - \lambda}$

$= e^{\lambda(e^{\theta}-1)}$

$Var(X) = E(X^2) - E(X)^2$

$= \lambda^2 + \lambda - (\lambda)^2$

$= \lambda$

Exercise 4.

$E(X) = \sum_{k=0}^{\infty} kP(X=k)$

$k = \sum_{j=0}^{k-1} 1$

$= \sum_{k=0}^{\infty} \sum_{j=0}^{k-1} P(X = k)$

$= \sum_{k=0}^{\infty} \sum_{k=j+1}^{\infty} P(X = k)$

$P(x > j) = \sum_{k=j+1}^{\infty} P(X = k)$

$P(X) = \sum_{j=0}^{\infty} P(X > j)$

Exercise 5.

Poisson moment generating function: $e^{\lambda(e^\theta - 1)}$

1st moment: $\frac{d}{d\theta} = \lambda e^{\theta + \lambda(e^\theta - 1)} = \lambda e^\theta e^{\lambda(e^\theta - 1)}$

$\theta = 0 => \lambda e^{0+\lambda(e^0-1)} = \lambda e^{\lambda(1-1)} = \lambda e^0 = \lambda$

2nd moment: $\frac{d^2}{d\theta^2} = \lambda \times 1e^\theta + \lambda e^\theta(\lambda e^{\theta+\lambda(e^\theta-1)}) = \lambda e^\theta + \lambda^2 e^{2\theta + \lambda(e^\theta - 1)}$

$\theta = 0 => \lambda e^0 + \lambda^2 e^{0+\lambda(e^0-1)} = \lambda + \lambda^2$

Exercise 6.

$E(X + 1) = Var(X)$

$E(-2X + 8) = 4Var(X)$

$E(X) + 1 = Var(X)$

$E(-2X) + 8 = 4Var(X)$

$-2E(X) + 8 = 4Var(X)$

$-2E(X) + 8 = 4(E(X) + 1)$

$-2E(X) + 8 = 4E(X) + 4$

$4 = 6E(X)$

$E(X) = \frac{2}{3}$

$Var(X) = \frac{2}{3} + 1 = \frac{5}{3}$

Exercise 7.

Pareto distribution PDF: $f(x) = \frac{\theta}{x^{\theta+1}}$ for $X > 1$ (=0 elsewhere) $\theta > 0$

$E(X) = \int_1^\infty xf(x)\,dx$

$E(X) = \int_1^\infty x\frac{\theta}{x^{\theta+1}}\,dx$

$= \theta\int_1^\infty \frac{x}{x^\theta x}\,dx$

$= \theta\int_1^\infty \frac{1}{x^\theta}\,dx$

$= \theta\int_1^\infty x^{-\theta}\,dx$

$\theta[\frac{1}{1-\theta}x^{1-\theta}]$

$\frac{\theta}{1-\theta}x^{1-\theta}$

The expected value is finite for $\theta > 1$

$Var(X) = E(X^2) - (E(X))^2$

$E(X^2) = \int_1^\infty x^2\frac{\theta}{x^{\theta+1}}\,dx$

$= \theta\int_1^\infty \frac{x}{x^\theta}\,dx$

$= \theta\int_1^\infty x^{1-\theta}\,dx$

$= \theta[\frac{1}{2-\theta}x^{2-\theta}]$

$= \frac{\theta}{2-\theta}x^{2-\theta}$

$Var(X) = \frac{\theta}{2-\theta}x^{2-\theta} - (\frac{\theta}{1-\theta}x^{1-\theta})^2$

$= \frac{\theta}{2-\theta}x^{2-\theta} - \frac{\theta^2}{(1-\theta)^2}x^{2-2\theta}$

$= \frac{\theta}{2-\theta}x^{2-\theta} - \frac{\theta}{1-2\theta+\theta^2}x^{2-2\theta}$

The variance is finite for $\theta > 2$

Exercise 8.

- trying to make spheres radius 1
- only can make sphere of radius R, R - unif$(1-\epsilon, 1+\epsilon)$
- PDF: $f(r) = \frac{1}{2\epsilon}$ for $1-\epsilon < r < 1+\epsilon$
- volume of sphere $= \frac{3}{4}\pi R^3$

(a) $P(V > \frac{3}{4}\pi) = P(R > 1)$

$P(R > 1) = \frac{1+\epsilon-1}{2\epsilon} = \frac{\epsilon}{2\epsilon} = \frac{1}{2}$

$P(V > \frac{3}{4}) = \frac{1}{2}$

(b) $E(V) = E(\frac{3}{4}\pi R^3)$

$E(\frac{4}{3}\pi R^3) = \frac{4}{3}\pi E(R^3)$

$E(r^3) = \int_{1+\epsilon}^{1-\epsilon} r^3 f(r)\, dr$

$= \int_{1-\epsilon}^{1+\epsilon} r^3 \frac{1}{2\epsilon}\, dr$

$= \frac{1}{2\epsilon} \int_{1-\epsilon}^{1+\epsilon} r^3\, dr$

$= \frac{1}{2\epsilon}[\frac{1}{4}r^4|_{1-\epsilon}^{1+\epsilon}]$

$= \frac{1}{2\epsilon}(\frac{1}{4}(1+\epsilon)^4 - \frac{1}{4}(1-\epsilon)^4)$

$= \frac{1}{8\epsilon}((1+\epsilon)^4 - (1-\epsilon)^4)$

$= \frac{1}{8\epsilon}(1 + 4\epsilon + 6\epsilon^2 + 4\epsilon^3 + \epsilon^4 - (1 - 4\epsilon + 6\epsilon^2 - 4\epsilon^3 + \epsilon^4))$

$= \frac{1}{8\epsilon}(8\epsilon + 8\epsilon^3)$

$= 1 + \epsilon^2$

$E(V) = \frac{4}{3}\pi(1+\epsilon^2)$

(c) expected error: $E(V) - \frac{4}{3}\pi$

$= \frac{4}{3}\pi(1+\epsilon^2) - \frac{4}{3}\pi$

$= \frac{4}{3}\pi + \frac{4}{3}\pi\epsilon^2 - \frac{4}{3}\pi$

$= \frac{4}{3}\pi\epsilon^2$

Since $\epsilon > 0$, the term $\epsilon^2 > 0$, which implies that the expected error is always positive.

Exercise 9.

- $x - Exp(\lambda)$ for some fixed $\lambda > 0$
- $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$, 0 elsewhere

$\lambda E(cos(\theta X)) = E(1 - \frac{(\theta x)^2}{2!} + \frac{(\theta x)^4}{4!} - \frac{(\theta x)^6}{6!} + -...)$

$= E(1) - E(\frac{(\theta x)^2}{2!} + E(\frac{(\theta x)^4}{4!} - E(\frac{(\theta x)^6}{6!} + -...$

$= 1 - \frac{\theta^2 E(X^2)}{2!} + \frac{\theta^4 E(X^4)}{4!} - \frac{\theta^6 E(X^6)}{6!} + -...$

$= 1 - \frac{\theta^2 \frac{2!}{\lambda^2}}{2!} + \frac{\theta^4 \frac{4!}{\lambda^4}}{4!} - \frac{\theta^6 \frac{6!}{\lambda^6}}{6!} + -...$

$= 1 - \frac{\theta^2 2!}{2! \lambda^2} + \frac{\theta^4 4!}{4! \lambda^4} - \frac{\theta^6 6!}{6! \lambda^6} + -...$

$= 1 - \frac{\theta^2}{\lambda^2} + \frac{\theta^4}{\lambda^4} - \frac{\theta^6}{\lambda^6} + -...$

$E(cos(\theta X)) = \frac{1}{1+\frac{\theta^2}{\lambda^2}}$

$E(cos(\theta X) = \frac{\lambda^2}{\lambda^2+\theta^2}$

Exercise 10.

- $X - unif(a, b)$
- uniformly distributed on the interval [a, b] if
- $f(x) = \frac{1}{b-a}$ for $a \leq x \leq b$, 0 elsewhere

$E(x) = \mu = \int_a^b x \frac{1}{b-a} \, dx$

$= \frac{1}{b-a} \int_a^b x \, dx$

$= \frac{1}{b-a} [\frac{1}{2} x^2 |_a^b]$

$= \frac{1}{b-a} [\frac{1}{2} b^2 - \frac{1}{2} a^2]$

$= \frac{1}{b-a} \frac{b^2-a^2}{2}$

$= \frac{b^2-a^2}{2(b-a)}$

$= \frac{(b-a)(b+a)}{2(b-a)}$

$= \frac{b+a}{2}$

$Var(X) = E(X^2) - E(X)^2$

$E(X^2) = \int_a^b x^2 \frac{1}{b-a}\, dx$
$= \frac{1}{b-a} \int_a^b x^2\, dx$
$= \frac{1}{b-a} [\frac{1}{3} x^3|_a^b]$
$= \frac{1}{b-a} (\frac{1}{3} b^3 - \frac{1}{3} a^3)$
$= \frac{1}{b-a} \frac{b^3-a^3}{3}$
$= \frac{(a^2+ab+b^2)(b-a)}{3(b-a)}$
$= \frac{a^2+ab+b^2}{3}$

$Var(X) = \frac{a^2+ab+b^2}{3} - (\frac{b+a}{2})^2$
$= \frac{a^2+ab+b^2}{3} - \frac{a^2+2ab+b^2}{4}$
$= \frac{4a^2+4ab+4b^2-3a^2-6ab-3b^2}{12}$
$= \frac{a^2+-2ab+b^2}{12}$
$= \frac{(a-b)^2}{12}$

Exercise 11.

- symmetric: $f(c-x) = f(c+x)$ for every real value of x
- X - continuous random variable with a PDF f(x) that is symmetric about c
- show: $E((X-c)^3) = 0, \mu_3 = 0$

$E((x-c)^3) = \int_{-\infty}^{\infty} (x-c)^3 f(x)\, dx$

$$u = x - c, x = u + c$$

$= \int_{-\infty}^{\infty} (u)^3 f(u+c)\, du$

Since the PDF is symmetric for c, we know that $f(c+u) = f(c-u)$

$$\int_{-\infty}^{\infty} (u)^3 f(c+u)\, du = \int_{-\infty}^{\infty} (u)^3 f(c-u)\, du$$

The function has the properties that $(-u)^3 = -u^3$, which is characteristic of an odd function and $f(c-u) = f(c+u)$, which is characteristic of an even function. Therefore, $(-u)^3 f(c-u) = -u^3 f(c+u)$, which characteristic of an odd function, so the overall function is odd.

If the function is odd and the upper and lower limits are opposite values (∞ and $-\infty$ are opposite values, and the function as we just established is odd) then the integral equals 0.

$$= \int_{-\infty}^{\infty} u^3 f(c+u)\, du = 0$$

So,

$$E((X-c)^3) = 0$$

Skewness: $\mu_3 = E((\frac{x-\mu}{\sigma})^3)$

$= \frac{E((x-\mu)^3)}{\sigma^3}$

Since the function is symmetric the mean, μ, must be equal to 0. But let's let $c = \mu$, since c is a constant value for every real value of x and μ is a real value of X. So $E((X-\mu)^3) = E((X-c)^3) = 0$

$= \frac{E((x-\mu)^3)}{\sigma^3} = \frac{E((x-c)^3)}{\sigma^3} = \frac{0}{\sigma^3} = 0$

Therefore, for a symmetric PDF, $E((X-c)^3) = 0$, and therefore the coefficient of skewness must also be 0.

Exercise 12.

$f(x) = \begin{cases} \frac{1}{\beta^2} x e^{-\frac{1}{2}(\frac{x}{\beta})^2}, x > 0 \\ 0, x \leq 0 \end{cases}$

$F(x) = \int_{-\infty}^{\infty} f(x)\, dx = \int_{-\infty}^{0} 0\, dx + \int_{0}^{\infty} \frac{1}{\beta^2} x^{-\frac{1}{2}(\frac{x}{\beta})^2}\, dx$

$= \int_0^x \frac{1}{\beta^2} x^{-\frac{x^2}{2\beta^2}}\, dx$

$$u = -x^2, du = -2xdx$$

$= \int_0^x \frac{1}{\beta^2} \frac{1}{-2} e^{\frac{u}{2\beta^2}}\, du$

$= -\frac{1}{2\beta^2} \int_0^x e^{\frac{u}{2\beta^2}}\, du$

$= -\frac{1}{2\beta^2} [2\beta^2 e^{\frac{u}{2\beta^2}} |_0^x]$

$= -e^{\frac{u}{2\beta^2}}|_0^x$

$= -e^{-\frac{x^2}{2\beta^2}}|_0^x$

$= -e^{-\frac{x^2}{2\beta^2}} - -e^0$

$= 1 - e^{-\frac{x^2}{2\beta^2}}$

$F(x) = \begin{cases} 1 - e^{-\frac{x^2}{2\beta^2}}, x > 0 \\ 0, x \leq 0 \end{cases}$

Exercise 13.

$F(x) = \frac{1}{1+e^{-(x-\alpha)/\beta}}$

$f(x) = \frac{d}{dx}(\frac{1}{1+e^{-(x-\alpha)/\beta}})$

$= -\frac{e^{-(x-\alpha)/\beta}(-\frac{1}{b})}{(1+e^{-(x-\alpha)/\beta})^2}$

$f(x) = \frac{e^{-(x-\alpha)/\beta}}{b(1+e^{-(x-\alpha)/\beta})^2}, -\infty < x < \infty$

Exercise 14.

(a)

- Supp(0,1)
- P(X = 0) = 0.2, P(X = 1) = 0.8

$M_x(\theta) = E(e^{\theta x}) = \sum_x e^{\theta x} P(X = x)$

$M_x(\theta) = e^{\theta 0} P(X = 0) + e^{\theta 1} P(X = 1)$

$= e^0 \times 0.2 + e^{\theta} \times 0.8$

$= 0.2 + 0.8e^{\theta}$

(b) Y - unif(0,1)

$f_y(y) = \begin{cases} 1, 0 \leq y \leq 1 \\ 0, otherwise \end{cases}$

Case 1: $\theta \neq 0$

$M_y(\theta) = E(e^{\theta y}) = \int_0^1 e^{\theta y} f_y(y)\, dy$

$= \int_0^1 e^{\theta y}\, dy$

$= \frac{1}{\theta} e^{\theta y}|_0^1$

$= \frac{1}{\theta} e^{\theta} - \frac{1}{\theta} = \frac{e^{\theta}-1}{\theta}$

Case 2: $\theta = 0$

$M(\theta) = E(e^{\theta y}) = E(e^{0 \times y}) = E(e^0) = 1$

(c) $M(\theta) = \frac{3}{4}e^{-\theta} + \frac{1}{8} + \frac{1}{8}e^{\theta}$

$Supp(-1, 0, 1)$ => corresponding to $e^{-\theta}, 1, e^{\theta}$

$P(X = -1) = \frac{3}{4}, P(X = 0) = \frac{1}{8}, P(X = 1) = \frac{1}{8}$

Exercise 15.

$$f(x) = \begin{cases} x^2, 0 \le x \le 1 \\ 1, 2 \le x \le \frac{8}{3} \\ 0, otherwise \end{cases}$$

$$f(x) = \begin{cases} 0, -\infty < x < 0 \\ x^2, 0 \le x \le 1 \\ 0, 1 < x < 2 \\ 1, 2 \le x \le \frac{8}{3} \\ 0, \frac{8}{3} < x < \infty \end{cases}$$

$F(x) = \int f(x) dx$

$$f(x) = \begin{cases} 0, -\infty < x < 0 \\ \int_0^x x^2\, dx, 0 \le x \le 1 \\ \int_0^1 x^2\, dx, 1 < x < 2 \\ \int_0^1 x^2\, dx + \int_2^x 1\, dx, 2 \le x \le \frac{8}{3} \\ 1, \frac{8}{3} < x < \infty \end{cases}$$

(because all probability mass is accounted for)

$$F(x) = \begin{cases} 0, -\infty < x < 0 \\ \frac{1}{3}x^3, 0 \leq x \leq 1 \\ \frac{1}{3}, 1 < x < 2 \\ \frac{1}{3} + (x-2), 2 \leq x \leq \frac{8}{3} \\ 1, \frac{8}{3} < x < \infty \end{cases}$$

Exercise 16.

U - uniform(0, 1)

$f_u(u) = \begin{cases} 1, 0 \leq u \leq 1 \\ 0, otherwise \end{cases}$

$E(U) = \int_0^1 uf(u)\,du$

$= \int_0^1 u \times 1\,du$

$= \frac{1}{2}u^2|_0^1$

$= \frac{1}{2} - 0$

$= \frac{1}{2}$

$Var(U) = E(U^2) - E(U)^2$

$E(U^2) = \int_0^1 u^2 \times 1\,du$

$= \frac{1}{3}|_0^1$

$= \frac{1}{3}$

$Var(U) = \frac{1}{3} - (\frac{1}{2})^2$

$Var(U) = \frac{1}{3} - \frac{1}{4}$

$Var(U) = \frac{1}{12}$

$X = (b-a)U + a$

$E(X) = E((b-a)U + a)$

$= (b-a)E(U) + a$

$= (b-a)\frac{1}{2} + a$

$= \frac{b-a}{2} + a$

$= \frac{b+a}{2}$

$Var(X) = Var((b-a)U + a)$

$= (b-a)^2 Var(U)$

$= (b=a)^2 \frac{1}{12}$

$= \frac{(b-a)^2}{12}$

Exercise 17.

$f(x) = xe^{x^2} I_{[0,1]}(x)$

Two conditions for a valid PDF:

1. Non-negativity: A valid PDF must satisfy $f(x) > 0$ for all x
2. Normalization: The total integral of the PDF over its support must equal 1

(1) The indicator function $I_{[0,1]}$ (x) is equal to 1 if $x \in [0,1]$ and 0 otherwise.

So, $f(x) = xe^{x^2}$ for $0 \leq x \leq 1$, and $f(x) = 0$ for $x \notin [0,1]$

For $0 \leq x \leq 1$, the function xe^{x^2} is non-negative, as both $x \geq 0$ and $e^{x^2} > 0$. So, $f(x) \geq 0$ for all x.

This condition is satisfied

(2) $= \int_0^1 xe^{x^2}\, dx$

$$u = x^2, du = 2x\, dx$$

$= \int_0^1 \frac{1}{2} e^u\, du$

$= \frac{1}{2} e^u|_0^1$

$= \frac{1}{2}e^{x^2}|_0^1$

$= \frac{1}{2}e^1 - \frac{1}{2}e^0$

$= \frac{1}{2}e - \frac{1}{2}$

$= \frac{e-1}{2}$

Turning into valid PDF:

$f_n(x) = \frac{2}{e-1}xe^{x^2}I_{[0,1]}(x)$

This new function now satisfies both conditions for being a valid PDF.